

Project 2

September 11, 2026 by 5 p.m.

Complete each of the following problems. To submit this project, you must type your solutions using your preferred software (e.g., Word, LaTeX, or Typst) and submit your work as a PDF. For each problem, you must show your work in order to receive credit. No credit will be received if you do not show your work. You may include scans of handwritten work, but the final solution must be typed.

A food science research team is investigating the microbial growth of E. coli. The bacteria cultures are prepared from the same initial stock and plated onto identical growth media. Each culture is placed in its own incubation unit. Although the incubation units are set to the optimal growth temperature of 37°C , variation of each machine means that the temperature, T , is a random variable. A tolerance of α is set so that each machine guarantees that the temperature of each machine does not go below $(37 - \alpha)^{\circ}\text{C}$ or above $37 + \alpha^{\circ}\text{C}$. It is reasonable to assume that the temperature of each machine is held constant.

$$f_T(t) = c \cdot \left(1 - \frac{(t - 37)^2}{\alpha}\right); \quad 37 - \alpha \leq t \leq 37 + \alpha$$

1. Find the value of c that produces a valid pdf for temperature, for any given α .
2. Compute $E[X]$.
3. Compute $V[X]$.
4. Your colleague decides to use Fahrenheit in their report, which is computed as $^{\circ}\text{F} = \frac{9}{5}^{\circ}\text{C} + 32$.

It was established that E. coli doubles approximately every 20 minutes when grown under optimal conditions. Let R be the random variable denoting growth rate.

$$E_{R|T=t}[R] = \frac{\ln 2}{20} - \beta(t - 37)^2$$